

RFP — Enshrined Liquid Staking on Flare (eFLR)

A design document for a protocol-owned liquid staking layer: what it would have to contain, which decisions it forces, and what would kill it.

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Disclosure: the author holds FLR and sFLR, runs LP positions on Spectra with sFLR as underlying, has had occasional contact with the Spectra team, and has an interest in duration markets on Flare existing. Section 6 cuts against the sFLR position; Section 9 runs with the duration interest. Weigh accordingly.

"FIP.17" is used below as a placeholder. The next FIP number is the Foundation's to assign. "eFLR" (enshrined FLR) is likewise a working name.

1. Executive Summary

Flare limits staking concentration by counting boxes: four validator nodes per registered entity, 300 million FLR per node, a 15× delegation factor on a 1 million FLR self-bond. The limit is enforced by a C-chain registry, not by the P-chain, and it constrains the entity that runs validators. It cannot see the pool that delegates to them. The largest pool on the network holds 2.384 billion FLR, 11.08% of active stake, and is subject to no cap of any kind. The September 2026 Sceptre case showed what a rational operator does when the permitted route costs 20% of rewards and the alternative is a second registration: it registers.

A percentage cap per entity, floating with the network, is the first fix and belongs in FIP.02/FIP.05, not here. This document is about the second problem, the one a percentage cap does not touch. Capital that is capped on one route moves to another. A pool that may not run its own boxes delegates to strangers and buys them back with fee rebates. A proxy with kickbacks breaks no rule on the books and every intention behind them, and governance can chase it for years. The alternative is to stop regulating the pool and to own it: a protocol-issued liquid staking token, eFLR, whose delegation is routed by a published formula across the validator set, with a per-entity cap the chain enforces because the chain is the delegator. The router does not replace competition among validators; it moves the competition from access to pool operators to measured performance, and every entity competes for the same delegation on the same terms.

The precedent is FIP.16. Faced with MEV, an extraction the network could not police, Flare did not write a rule against extractors. It moved block building to a designated builder and routed the proceeds to FIRE. Private liquid staking is extraction of a different kind, of validator selection and of custody over pooled capital, and the same move is available.

This RFP lays out what an enshrined layer would need: the settlement problem (a C-chain contract cannot sign P-chain transactions, and that gap is where Sceptre's four keys live), the routing formula and its parameters, the fee and reserve design, the governance envelope, composability, and a migration path for the pools that exist today. It presents three scopes, not one: an enshrined router that existing pools must use, an enshrined token, or both. It then lists the ways it fails, starting with the strongest: an enshrined pool is itself the largest pool on the network, and its neutrality is exactly as good as the formula and the keys behind it.

The document is a request for proposal, not a specification. Numbers are proposals with reasons. The falsification tests in Section 11 carry dates.

2. The Problem: Capacity Is Not Independence

2.1 What the current rules measure

FIP.05 (September 2023) set the staking parameters that still govern: minimum self-bond 1 million FLR, delegation factor 15×, maximum 200 million FLR per node, up to four validators per infrastructure entity, uptime above 80% to earn rewards. The FIP.16 hard fork of 14 July 2026 raised the node maximum to 300 million. The entity ceiling is therefore 1.2 billion FLR.

Three things about that ceiling. It is absolute on a network whose active stake went from roughly 11 billion in March 2026 to 21.5 billion by September; a fixed number in a doubling system is a limit on growth wearing the costume of a limit on concentration. It is enforced by the entity registry on the C-chain, not by the P-chain, which FIP.05 says in plain words. And it measures the wrong layer. A node is a validator with its own stake; an entity is the registration that may run four of them; a pool is a contract that delegates to nodes it does or does not own. The rule constrains entities. The concentration sits in pools.

The one percentage that does exist, the 5% vote-power cap per validator above which delegators are diluted rather than refused, is slack: under a 300 million node cap no single node reaches 1.075 billion.

2.2 What FIP.16 changed

FIP.16 (April 2026) did four things that matter here. It cut inflation from 5% to 3%. It set a mandatory minimum 20% entity fee on everything staked or delegated to an entity. It raised the weight of P-chain stake to 5× C-chain delegation in the signing

weight that governs FTSO, FDC and block-latency rewards, which pushes capital from liquid WFLR delegation into locked P-chain stake. And it created FIRE, the Flare Income Reinvestment Entity, with a mandate to capture fees and MEV and a roadmap that moves block building to a FIRE-designated builder in three stages.

Put together: the network made third-party delegation more expensive (20% floor), made P-chain staking more rewarding ($5\times$ weight), and left the enforcement of the entity limit to a registry that counts addresses. A liquid staking pool above 1.2 billion FLR now pays 20% on everything above the ceiling unless it finds a way around the ceiling. Sceptre found one. On the author's estimate the avoided fee was on the order of 30 million FLR a year, more than the pool's entire service-fee revenue.

2.3 The proxy inevitability

Assume governance fixes the count. Suppose FIP.02 is amended, the Management Group chills the second identity, and a binding 5% share of active stake per entity is written into the reward script. What does a 2.4 billion FLR pool do?

It delegates the excess to independent validators, and it negotiates. The FIP.16 floor guarantees those validators 20% of rewards; a side agreement returns part of it. The P-chain sees distinct NodeIDs. The registry sees distinct entities. The reward script sees a diversified delegation. The capital is as concentrated as before, and its concentration is now invisible. Nothing in this is illegal, and nothing in it is what FIP.05 or FIP.16 meant.

This is not a hypothetical about Flare. It is the observed equilibrium of every proof-of-stake network that left liquid staking to the market: Lido reached roughly a third of Ethereum's stake through exactly this structure, a pool that owns no validators and selects all of them. A cap on the entity does not prevent it. A cap on the pool would, but the pool is a contract the Management Group cannot reach.

2.4 The FIP.16 precedent, stated precisely

The analogy that carries this document needs to be exact, because a loose version of it is wrong.

FIP.16 did not "ban MEV". It observed that MEV is a rational response to a protocol that leaves transaction ordering to whoever proposes the block, concluded that no rule against extractors would hold, and changed the protocol so that ordering is performed by a designated builder under a published mandate, with proceeds routed to FIRE. The extraction was internalized, not prohibited. The builder is initially the Foundation; Stage 2 moves the builder into Flare Confidential Compute; Stage 3 makes builder and proposer one protocol role.

The parallel to liquid staking is one-to-one. Validator selection by a private pool is rational under a protocol that leaves delegation to whoever holds the deposits. No rule against proxy delegation will hold. The alternative is a protocol that performs delegation itself, under a published formula, with the pool's custody moved from

private keys to protocol keys. Whether that is worth its costs is the question this document exists to ask.

3. Precedents and Their Verdicts

Three networks have faced this and answered differently. The answers are the design space.

Polkadot: nomination pools (2022). Protocol-native staking pools, in runtime, with no liquid token. Anyone may create a pool; members pool stake under one nominator account. It solved the participation problem (minimum 1 DOT) and did nothing about concentration, because pool operators still choose validators. Lesson: enshrining the pool without enshrining selection changes custody, not concentration.

Cosmos Hub: Liquid Staking Module (2023). Not an enshrined token but an enshrined constraint on private ones. Liquid staking providers may hold at most 25% of all staked ATOM (governance-adjustable), and every validator must self-bond at 1:250 against the liquid stake delegated to it. Private LSTs remain; the protocol caps their aggregate and forces validator skin in the game. Lesson: a protocol can bound pools without owning them, if the chain can distinguish pooled delegation from direct delegation.

Ethereum: the road not taken (2023-2025). Enshrined LSTs were argued in earnest by Ethereum Foundation researchers, first as a two-tier staking model, then as rainbow staking (heavy and light services, protocol-enforced self-limits for pools). None has shipped. The objections are the ones this document must answer: a protocol that selects validators has moved the political question into protocol parameters; a protocol token competes with private ones but does not remove them; and enshrinement adds consensus-level surface for a problem that might be solved by capping issuance. Lesson: enshrinement is a minority path, and its advocates lost the argument on the largest network. Flare's case has to be made on Flare's specifics, not on Ethereum's.

The specifics are favourable in two ways. Flare's validator set is small (179 active), and its validators are already FTSO data providers whose performance the protocol measures every reward epoch on-chain. The information an algorithmic router needs is already being computed. And Flare has already crossed the line Ethereum would not: FIP.16 designated a builder. A network that has enshrined block building has fewer principled objections left to enshrining delegation.

4. Scope Options

The sparring that produced this document started from "the protocol issues eFLR". That is one of three scopes, and the cheapest one is not it.

Scope A — Enshrined router, private tokens. The protocol builds a delegation router: a contract that accepts delegation intent from any pool and executes it across the validator set by formula, with per-entity caps. Private LSTs continue to exist and to issue their own tokens, but above a threshold share of network stake they are required to delegate through the router (the Cosmos LSM pattern, applied to routing rather than to a global cap). Solves selection and concentration; leaves custody with the pools.

Scope B — Enshrined token, protocol custody. The protocol issues eFLR against deposits, holds the capital under protocol keys, and delegates by formula. Private tokens compete on the market. Solves custody and selection for the capital that migrates; does nothing for capital that stays in private pools unless Scope A is also present.

Scope C — Both. The protocol issues eFLR and requires every pool above the threshold to route through the same router eFLR uses. Private pools become front-ends on protocol rails; their remaining product is distribution, integration and yield strategy on top of the base rate.

The recommendation of this RFP is C, staged: A first, because it can ship without solving the settlement problem in Section 5 and immediately bounds the pool that exists today; B on the FIP.16 Stage 3 consensus change, because that change already moves where staked funds reside. A alone leaves the four-keys problem in place. B alone leaves Sceptre's 2.4 billion where it is.

4.1 Decision matrix

A recommendation without the criteria behind it is an opinion. The criteria, as this RFP weighs them:

Criterion	Scope A (router)	Scope B (token)	Scope C (both)
Bounds the pool that exists today	Yes, at the routing layer	No	Yes
Removes operator custody risk	No	For migrated capital	For migrated capital; private pools keep their keys
Consensus change required	No	Yes (§5.2 ii) or FCC custody (§5.2 i)	Same as B
Governance surface added	Router parameters	Router parameters + custody envelope	Same as B
Time to Songbird	One to two quarters	Tied to Stage 3	A first, B on Stage 3
Political cost	Private pools lose selection, keep fees	Foundation competes with sFLR	Both
Regulatory exposure (§11.10)	Low: no new asset	Higher: protocol-issued yield-bearing asset	Same as B

Two decision conditions, stated as proposals with dates the Foundation should replace with its own. If by the end of Q1 2027 no contract-originated delegation primitive is specified on the Stage 3 path, decide A, ship it on Songbird, and treat B as closed until Stage 3 is scheduled. If a primitive is specified by then and running on Coston2 by Q3 2027, decide C and sequence B behind two Songbird reward-epoch cycles. What this RFP cannot supply is the cost side: engineering effort for the primitive, FCC enclave provisioning, audit budget. Those numbers belong to the core team and are requested in Section 10.

5. The Settlement Problem

This is the section a reviewer from the core team will read first, and it is the one where the sparring draft was silent.

5.1 Two chains, two delegations

Flare inherits Avalanche's architecture: a P-chain that holds validator stake and delegation, and a C-chain that runs the EVM. Staking rewards for P-chain stake are computed off-chain by a public script and distributed by C-chain contracts through a mirror (PChainStakeMirror). FTSO delegation is a different mechanism entirely: WFLR on the C-chain, delegated by percentage to a data provider, no lock, rewards every 3.5 days. FIP.16 unified the weight both carry in the FSP protocols and set P-chain stake at $5\times$.

A liquid staking pool on Flare has to do both. Sceptre's contract holds deposits on the C-chain, delegates WFLR to FTSO providers, and moves a portion to the P-chain to stake against validators. The second step is the problem. A C-chain contract cannot construct or sign a P-chain transaction. Somebody has to export FLR from C to P, issue `addDelegator` or `addValidator`, and later import it back. That somebody holds keys. In Sceptre's case, four accounts with `DEPOSIT` and `WITHDRAW` roles, granted on 15 July 2026, that moved roughly 140 million FLR out of the pool into self-bonds with no timelock, no cap and no notice.

The four keys are not a Sceptre design flaw. They are the only way a C-chain pool can reach the P-chain today. Every liquid staking token on Flare has a version of them.

5.2 Three ways to close the gap

(i) Protocol-managed keys in Flare Confidential Compute. FIP.16 Stage 2 already plans to run the block builder inside FCC. The same enclave model can hold the P-chain keys of the eFLR pool: keys generated inside a TEE, never exported, executing a fixed policy (export, delegate per router output, import at maturity) against attested inputs. This is the author's earlier "Ulysses at the mast" proposal applied to staking rather than treasury. It is buildable on the FCC roadmap and it does not change consensus. Its weakness is the enclave image: whoever can replace

it controls the pool, so the attestation and upgrade path for the image is the actual custody question.

(ii) A P-chain primitive for contract-originated delegation. The P-chain recognises a delegation whose owner is a C-chain contract address, with the mirror running in reverse: the contract emits intent, the P-chain executes it, the mirror proves execution. This is the clean answer. It is also a consensus change. FIP.16 Stage 3 says in terms that "the technical implementation of how exactly staking is handled, where the funds reside, and concrete staking processes will change in accordance with the consensus implementation changes." That sentence is the window. An enshrined pool designed now can ride that change; designed later it will have to be retrofitted.

(iii) Restrict eFLR to C-chain delegation. A pool that only delegates WFLR to FTSO providers needs no P-chain keys at all. It also earns at $1\times$ weight against $5\times$ for P-chain stake, and it does nothing for validator concentration, because it never touches validators. This is not a liquid staking token; it is a delegation wrapper, and it is mentioned so that it can be ruled out.

The RFP asks the core team one question above all others: is (ii) on the Stage 3 path, and if not, what would it cost to put it there? Everything in Section 7 assumes the answer to (i) or (ii) is yes.

Between (i) and (ii) the RFP takes a position rather than leaving it as a menu. The enclave route has a single point of failure in the image and its attestation chain, and a compromise there is a loss of the pool with no recovery path except a hard fork (§11.11). The primitive route has consensus risk, which is bounded by Songbird testing and by the fact that Stage 3 already opens the consensus for changes of this kind. A consensus change tested on Songbird is a risk the network has taken before; an enclave holding a multi-billion FLR pool is not. The primitive is the target; the enclave is acceptable only as a bridge with a published sunset. Appendix A sets out the minimum each option would have to specify for the core team to answer with more than "possible".

5.3 Liquidity: the exit queue

P-chain delegation locks for a chosen period, minimum two weeks, and cannot be withdrawn early. A liquid token over locked capital needs either an instant-redemption reserve (Sceptre's model, and the reserve the four keys drew down) or an exit queue with a redemption delay, or both. The sparring draft assumed instant, zero-fee redemption during a governance timelock; that is not achievable against locked P-chain stake without a reserve large enough to honour it.

The proposal: a ladder delegation schedule (delegations staggered so that a fixed share matures every epoch), an exit queue served in order from maturing delegations, and an instant-redemption reserve whose minimum size is a contract invariant, not an administrator's discretion. The ragequit right in Section 8 is the right to enter the queue at the head, not the right to instant exit.

6. Architecture

6.1 Deposit and token

Users deposit FLR (or WFLR) and receive eFLR. Two accounting models exist. A rebasing token (balance grows) is simpler to reason about and breaks most DeFi integrations. A value-accruing token (balance fixed, exchange rate to FLR rises, sFLR's model) is what every lending market, AMM and yield-stripping protocol on Flare already integrates. eFLR should be value-accruing. This is not a small choice: it determines whether Spectra, Kinetic and Enosys can list eFLR on day one by cloning their sFLR configuration.

6.2 Capital sources

Two inflows. User deposits, which are the point. And, as an option the RFP puts on the table rather than assumes, protocol-owned capital: FIRE's mandate already ranks "rewards to FLR validators and stakers" as its second priority, and FIRE's captured FLR could be deposited into eFLR rather than distributed. The effect would be to seed the pool and to give FIRE a yield-bearing FLR position. Two cautions. FIRE's assets are mixed (stablecoins, FXRP, FLR), so only the FLR leg qualifies. And a pool part-owned by the Foundation's revenue entity is a pool whose neutrality will be questioned; Section 8 addresses whether the governance envelope can carry that.

The "deflationary flywheel" claim from the sparring draft is dropped. Locking FLR in a pool changes float, not supply. The fee in Section 7.2 burns a fraction of yield that is small against 3% inflation. eFLR is a supply sink in the FIP.16 sense (excluded from the inflatable base while locked) and nothing more, and the document should not say more.

6.3 The delegation router

The router is the mechanism that replaces Sceptre's validator analysts. Every reward epoch (3.5 days) it computes a target allocation of the pool's stake across eligible entities and issues the delegation instructions needed to move toward it, subject to lock schedules.

Target allocation per entity i :

$$A_i = W \cdot (T_i \cdot Y_i) / \sum_j (T_j \cdot Y_j) \quad \text{capped at } C_{\max}$$

where W is the pool's delegable stake, T is a trust weight, Y is a net-yield weight, and $C_{\max}$ is the per-entity cap. Surplus above any entity's cap redistributes to the next by the same weights. Delegation is per entity, summed over that entity's registered nodes, because the entity is what the registry knows and the node is what the P-chain knows; the router must reconcile both.

Three terms, used consistently from here:

Term	Definition	Known to	Role in the router
Node	One validator with its own NodeID and stake	P-chain	Unit of delegation execution
Entity	Registered identity, one to four nodes	C-chain registry	Unit of scoring (T, Y) and of the cap C_max
Operator	The organisation behind one or more entities	Nobody on-chain	Target of the correlation rule (6.8)

The router scores and caps entities, executes against nodes, and applies the correlation rule at entity level: an entity that spreads four nodes across four hosts does not thereby escape a penalty it shares with another entity on any one of them.

6.4 Eligibility (hard filters)

An entity is eligible if, over the evaluation window, it has met the FIP.05 uptime requirement (80%; the RFP proposes the router use a stricter 95%), has been rewarded by the FSP for FTSO data provision in every epoch of the window (which is what "good enough prices" means in FIP.05 terms and is already computed on-chain), holds no active chill under FIP.02, and has a self-bond meeting the P-chain minimum. There is no slashing on Flare's P-chain; "slashing history" in the sparring spec is replaced by chill history, which is the network's actual sanction record.

6.5 Trust weight T_i

New entities enter at $T = 0.5$ and rise linearly to $T = 1.0$ over 50 reward epochs (about six months) of continuous eligibility. The purpose is to make Sybil entry expensive in time, since a 1 million FLR self-bond does not make it expensive in capital. A window of 30 epochs (about 105 days) governs penalties: each epoch of ineligibility inside the window deducts 0.2 from T, floored at 0; eligibility restores T by 0.05 per clean epoch. An entity that goes dark for one epoch loses a fifth of its weight for four epochs and is back at full weight within a month. An entity that is chilled loses eligibility for the chill's duration and re-enters at $T = 0.5$. No lifetime exclusions; the router is not a court.

The obvious attack is identity splitting: an operator at C_max registers a second entity, waits out the ramp, and after six months collects delegation under two names. Two things bound it, and neither is the ramp. FIP.05's delegation factor caps total stake on a node at 15× its self-bond, so a fresh entity with the 1 million FLR minimum can receive at most 14 million in delegation from anyone, router included; scaling the attack means scaling self-bond, which is capital the operator has to lock. And the cap in 6.7 is computed on the entity's share of network stake, so a second entity that shares infrastructure with the first shares its cap under 6.8. A reviewer proposed tying the ramp to self-bond size directly (T rising more slowly below a 5 million FLR bond). The RFP does not adopt it: the delegation factor already prices Sybil entry in capital, and a bond-weighted ramp would slow exactly the small honest

operators the baseline exists for. What the ramp buys is time, and time is the one input a well-funded attacker cannot buy.

6.6 Net-yield weight Y_i

Flare's validators are data providers, and their reward rate varies with data quality. A router that ignores that pays bad data providers to exist. Y_i is the entity's realised net reward rate to delegators over the trailing 10 epochs, divided by the network median:

$$Y_i = (\text{gross reward rate}_i \times (1 - \text{fee}_i)) / \text{median}_j(\text{gross reward rate}_j \times (1 - \text{fee}_j))$$

Fee is the entity's declared fee, floored at 20% by FIP.16. An entity charging 35% competes against one charging 20% on net yield and loses delegation to it; an entity whose FTSO submissions earn more competes on gross and wins. Y is clamped to [0.5, 1.5] and computed on a trailing ten-epoch mean, so that one exceptional epoch moves an entity's weight by at most a tenth of its excess and the router remains a baseline, not a momentum trader. Yield-hopping, where delegation chases last epoch's winner, is a failure mode of every reactive allocator, and the window and the clamp are the two instruments against it; if either proves too loose on Songbird, both are parameters.

This is the part of the design that answers the objection that an enshrined pool socialises the validator set. It does the opposite. Today delegation from the largest pool follows a relationship with the pool operator, and the terms of that relationship are not public. Under the router, delegation follows three numbers every entity can see and move: uptime, data quality, and fee. An entity that improves its FTSO feeds gains delegation next epoch from a delegator that cannot be lobbied. The competition is not dampened; it is made legible and open to every entity that meets the filters, including the ones that have no business-development function at all.

6.7 Cap $C_{\max}$

The cap is the point of the whole design and the parameter most likely to be fought over. The sparring draft proposed 3% of the pool per node. That is the wrong denominator. A pool that is 40% of the network with a 3% per-node cap still hands 1.2% of the network to a node in one instruction; and a cap on the pool's share says nothing about the entity's total share once its other delegators are counted.

The proposal: $C_{\max}$ is expressed as a share of total active network stake, per entity, inclusive of the entity's stake from all sources as the reward script already sees it, and set so that the router never pushes an entity across the binding percentage cap this RFP assumes governance will adopt in FIP.02/FIP.05. If that cap is 5%, the router's $C_{\max}$ is 5% minus the entity's non-router stake. The router then does with code what the Management Group cannot do with a forum: it makes the largest delegator on the network incapable of concentrating.

6.8 Correlation

Eight of Sceptre's nodes sat in one /24 on one host. A cap per entity does not see that two entities share a rack. The router should carry a correlation penalty on top of C_max: entities whose nodes share an autonomous system, hosting provider or /24 with another eligible entity share one cap between them for the shared portion. The data exists (Catenalytica already attributes hosts); whether it can be made oracle-grade is an open question in Section 10.

7. Economics

7.1 The base rate

The pool's gross yield is the stake-weighted average of what its delegated entities earn, less their fees (20% floor), less the protocol fee below. At today's parameters an sFLR holder sees roughly 7–8% gross on the pool, of which Sceptre takes 10% and the network's 3% inflation absorbs most of the rest in real terms. The author's estimate of Sceptre's numbers: about 228 million FLR a year of pool rewards on 2.384 billion, about 23 million of which is service fee.

eFLR's advantage over that is not primarily the fee. It is that the router pays the 20% floor on everything and delegates within the entity ceiling by construction. A private pool that runs its own validators can beat eFLR on yield, exactly as Sceptre did, by taking the self-bond rewards for itself. That advantage is real and it is the advantage the network wants to remove. eFLR does not win on yield; it wins on not being the counterparty that can take the reserve.

7.2 Protocol fee

A fee on pool rewards, proposed at 5% for discussion; Sceptre charges 10%, typical LST fees run 10–20%, and the right number is whatever covers the reserve and leaves eFLR's net rate above the private pools' without a self-bond subsidy. Two destinations, and the sparring draft's 50/50 split is replaced with something the numbers can carry.

The first destination is the instant-redemption reserve, until it reaches its invariant size (proposed: 5% of pool). Once the reserve is full, the fee goes to the second destination, and here the RFP has to name a conflict it created. FIRE is administered by the Foundation. A fee from eFLR to FIRE is, in practice, a fee to the Foundation, and it gives the Foundation a direct financial stake in eFLR's growth. Every parameter in Section 6 that makes eFLR more attractive to depositors (a looser uptime filter, a softer Y clamp, a higher cap) raises that fee. If the same Foundation also holds the custody keys, the party that sets the router's neutrality is paid for the router's size. That is not a hypothetical; it is the structure.

The fee's destination is therefore conditional. If the custody condition in Section 8 is met, and no single party including the Foundation can replace the enclave image or the upgrade key, the fee may go to FIRE under its existing mandate, because FIRE's priorities two through four are the ones an eFLR fee should fund. If the custody condition is not met, the fee is burned. Burn benefits every FLR holder equally and gives no party an interest in the pool's size. The RFP does not propose a separate "insurance fund". Flare's P-chain does not slash, so there is no principal loss to insure against from validator conduct. The arithmetic for the risks that remain is short: a 2 billion FLR pool at 7% gross yields about 140 million FLR a year, of which 5% is 7 million; an enclave compromise loses the pool. A fund that claims to cover a loss three hundred times its annual inflow is marketing, and "principal-protected" does not appear in this document. What a compromise does get is \$11.11.

7.3 Who earns what

The depositor earns the base rate net of 20% entity fees and 5% protocol fee, holds a token the counterparty cannot drain, and gives up the possibility of the above-market yield a self-validating pool can offer. The entity earns the 20% floor on router delegation, allocated by uptime, data quality and fee discipline rather than by relationship with a pool operator; small entities that perform get delegation they cannot market for today. The protocol earns a fee stream into FIRE and, more importantly, a validator set whose largest delegator is incapable of concentration by construction. The private pool operator loses the yield edge from self-bonding and the discretionary selection business, and keeps distribution, integrations and strategy layers.

7.4 What this does to the 20% floor

FIP.16's floor exists to make independent entities viable. A router that delegates by net yield puts every entity at the floor, because any fee above it loses delegation. That is the intended outcome and it should be said plainly: the router makes the floor the ceiling. If governance wants a market in entity fees above 20%, the Y weighting has to be softened. The RFP's position is that it should not be.

8. Governance Envelope

The sparring conversation's own objection: enshrining the pool moves the bottleneck from four private keys to whoever can change the contract. On Flare that is, today, the Foundation, through the governance timelock on every system contract. The FIRE precedent is instructive and not reassuring: FIRE is administered by the Foundation with a "negative governance" veto that requires 50% of the inflatable supply to vote, a threshold no vote has reached.

The envelope this RFP proposes has four parts, and the fourth is the one that matters.

Immutable core. Deposit, redemption, exit queue, the reserve invariant, and the cap formula are non-upgradable. If they are wrong, the pool is wound down and redeployed; it is not patched.

Parameter set under timelock. T ramp length, penalty size, Y clamp, uptime threshold, fee, reserve size, correlation rule: adjustable, with a 21-day on-chain timelock and public diff before effect.

Ragequit during timelock. Any pending parameter change opens a window in which eFLR holders may enter the exit queue at the head, ahead of ordinary redemptions, served from the reserve and maturing delegations. It is not instant exit; Section 5.3 explains why it cannot be. It is guaranteed priority.

Custody outside the Foundation. This is the part the sparring draft's dual-veto does not solve. A veto over parameters is worth little if the enclave image, the router's data feeds and the upgrade key are held by the same party. The four buckles from the author's FIRE essay apply verbatim: the upgrade key, the mandate governance, the enclave image, the input feeds. Each must have a documented holder, and for at least the enclave image and the upgrade key the holder must not be the Foundation alone. A 2-of-3 across Foundation, an elected entity committee (FIP.16's own joint-governance mechanism, Songbird-then-Flare election) and a time-delayed community key is the minimum. If the Foundation will not accept that, Scope A (router without protocol custody) is the honest fallback, and the four keys stay with the pools, bounded by the contract terms the earlier dispatch asked for.

Transition. A 2-of-3 that does not exist at launch is a promise, and Section 3 of the author's FIRE essay already said what promises are worth. The proposal is staged so that each stage is a precondition for the next, not a hope after it. At launch, on Songbird, the Foundation holds the keys alone, under the 21-day timelock and the ragequit right; this is acceptable because Songbird is the test and the amounts are SGB. Before any Flare deployment, the entity committee exists: elected by the mechanism FIP.16 §4.5.1 already specifies (a Songbird vote to shortlist eight, a Flare vote to seat four) but without FIP.16's precondition of a 50%-of-supply vote to unlock it, which has never been met and was designed not to be. Before the pool exceeds a threshold share of active stake (proposed: 10%), the community key exists: a time-delayed multisig whose signers are elected by eFLR holders, with the power to block a parameter change or an image replacement and no power to initiate either. If the second or third stage is not in place when its trigger arrives, deposits close until it is. Closing deposits is the enforcement; there is no other.

Deadlock. Three parties will disagree. The rule is that the status quo wins: any change to parameters or to the enclave image needs two of three, so one party can block, and a block is not a crisis, it is the design working. Where all three want a change and differ on the value, the most conservative proposal takes effect (the lowest cap, the strictest filter, the longest timelock), and the others may propose again after one reward epoch. Where the disagreement is about an emergency (a live exploit, an image with a known bug), the only emergency action available to any single party is to pause new deposits and new delegation instructions; existing

delegations mature on schedule and the exit queue keeps running. Nobody can move capital alone, in an emergency or otherwise. That is the whole point of the envelope, and an emergency power that could override it would be the four keys again.

9. Composability

eFLR is a base asset. Everything below it in the stack (fixed-term lending, options collateral, yield stripping) exists or is proposed elsewhere, and this section decides what the protocol builds and what it leaves.

The sparring draft proposed native PT/YT splitting in the protocol. This RFP recommends against it. Spectra already runs yield tokenisation on Flare against sFLR, with live pools across maturities, and the Fixed-Term Lending RFP already specifies PT-as-collateral on Spectra's rails. An enshrined splitter duplicates a live product, adds contract surface to the core, and forces the protocol to maintain a maturity calendar. What the protocol should do instead is make eFLR trivially strippable: value-accruing accounting, a clean exchange-rate oracle, no transfer hooks, no rebasing. Spectra then lists eFLR as it listed sFLR, and PT-eFLR becomes the collateral the lending RFP wants: a claim on protocol-custodied FLR with no operator key risk, which is the property that makes it institutional-grade rather than merely fixed-rate.

The author's conflict is on the table here: the recommendation favours a protocol the author uses and a market the author wants to exist. The counter-argument, that enshrined stripping removes a dependency on a third party, is real, and Section 10 keeps it open.

FAssets. The largest consumer of FLR collateral on the network is not a lending market; it is the FAssets system, where FLR sits in agents' collateral pools behind FXRP. eFLR in that role would replace an asset with operator key risk (sFLR) with one without it, which is an argument FAssets governance should hear, and it comes with a liquidity condition: an FAssets liquidation that receives eFLR has to be able to turn it into FLR faster than the exit queue allows. Two ways to meet that. A haircut on eFLR's collateral value sized to the secondary-market discount observed on Songbird under stress, which is the honest way and needs no new mechanism. Or a liquidation lane in the reserve, ahead of ordinary redemptions and behind ragequit, capped at a share of the reserve the invariant can spare. The RFP proposes the first and leaves the second to the FAssets team, and it does not propose collateral factors; those are FAssets governance's to set against data that does not yet exist.

10. Open Questions

Ordered by how much the answer changes the design.

1. **Is contract-originated P-chain delegation on the FIP.16 Stage 3 path?** If yes, Scope B is a Stage 3 deliverable. If no, Scope B depends on FCC-managed keys and the enclave image becomes the custody question.
2. **Can the router's inputs be made oracle-grade?** Uptime and FTSO reward data are on-chain; host and AS attribution are not. The correlation rule (6.8) needs a source the protocol can verify or it stays advisory.
3. **Does the registry refuse a fifth node?** The earlier dispatch could not confirm it. If the registry enforces the count, Scope A can bind pools at the registry; if not, the mirror service is the enforcement point and phase 3 is a prerequisite.
4. **What threshold triggers mandatory routing under Scope A?** Cosmos chose 25% of all stake for the aggregate of liquid providers. Flare's largest pool is at 11%. A per-pool threshold of 5% of active stake is the number consistent with the entity cap; the RFP asks whether it should be lower.
5. **Should FIRE seed the pool?** Section 6.2 puts the option forward and names the neutrality cost. The answer depends on Section 8's custody outcome.
6. **Native stripping or Spectra rails?** Section 9 recommends rails. The dependency argument against it deserves a written answer from the Foundation, not from this author.
7. **What happens to sFLR?** Three paths, and the RFP should be plain about which ones work. A soft path (eFLR launches, sFLR stays, a conversion at the snapshot exchange rate with a small bonus for early movers) is politically costless and, on the Cosmos and Ethereum evidence, does not move capital; holders stay where their integrations are. A hard path (Scope C: sFLR must delegate through the router above the threshold) removes the concentration and leaves Sceptre its 10% service fee, its front-end and its integrations, which is why it is the path a rational Sceptre should prefer to a percentage cap it has to route around. A third path, freezing or force-converting the sFLR contract, is excluded here and should be excluded in writing by the Foundation: it would establish that a protocol can seize a private contract's deposits by vote, and that precedent is worth more to an attacker than any pool. The honest statement is that without the hard path eFLR does not reach critical mass, and the decision between soft and hard is the Foundation's political decision, not a technical one.
8. **Songbird first?** Every consensus-touching FIP has shipped on Songbird first. The router (Scope A) can run on Songbird against SGB staking within a quarter of a decision; the token (Scope B) should not ship on Flare before it has run through at least two Songbird reward-epoch cycles with the exit queue under load.
9. **What does it cost?** Engineering effort for the P-chain primitive, FCC provisioning for the enclave route, audit scope for the immutable core, and the reserve's opportunity cost. This RFP has no basis for those numbers and does not invent them. The decision matrix in 4.1 is incomplete without them, and the

request is that the core team fill that column before the scope decision, not after.

10. **Will FAssets governance accept eFLR as pool collateral, and on what haircut?** Section 9 argues it should; the number depends on Songbird stress data that does not exist yet.
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11. Failure Modes

Hardest first. Before the list, who measures: every test below carries a date and a threshold, and a Foundation that measures its own project will find it succeeding. The proposal is that the metrics (pool share of active stake, share of pooled stake routed, entity count receiving router delegation, exit-queue depth, secondary-market discount) are published as time series by a party that does not hold keys, on the pattern Flare Metrics and Catenalytica already follow, quarterly, on the forum. Three consecutive missed quarterly thresholds open a review in which the three custody parties of Section 8 must, within one reward-epoch cycle, choose one of: adjust parameters, change scope, or wind down. Under Scope C "routed" means delegation that passed through the router from any pool, sFLR included; eFLR's own share is a separate metric and a weaker one.

11.1 The enshrined pool is the largest pool. If eFLR works it becomes 30–50% of active stake, and the largest delegator on the network is a contract whose parameters are set by governance and whose keys are held by whoever holds them. Every objection to Sceptre's 11% applies at three times the size. The design's defence is that the router cannot concentrate (6.7) and the keys cannot be used discretionarily (8). Both defences are exactly as strong as their implementation and no stronger. If the four buckles are not closed, this RFP has proposed a bigger Sceptre with a Foundation logo. Test: before mainnet, an independent review of the enclave image and key holders, published; if any single party can replace the image, do not ship Scope B.

11.2 The router becomes the politics. Every parameter in Section 6 is a decision about who gets delegation. Entities will lobby for the uptime threshold, the Y clamp, the correlation rule. The forum fight about Sceptre's identities becomes a permanent fight about T and Y. Mitigation: parameters under timelock with ragequit, so that a lost fight is an exit; and a rule that parameter changes take effect only at reward-epoch boundaries with a 21-day notice. This does not remove the politics. It gives it a schedule.

11.3 The proxy problem survives. A cap per entity inside the router does not stop a cartel of twenty entities from taking 60% of it. Section 6.8's correlation rule catches shared infrastructure; it does not catch shared ownership on separate infrastructure. The document should not claim otherwise. The honest statement: enshrinement removes the pool operator as the concentrating actor and leaves validator-level collusion where every proof-of-stake network leaves it, at the limit of what code can see. An identity-staking or later-established-breach slashing rule,

whose buildability on Flare's P-chain is unresolved, is the next instrument, and it is not in this RFP.

11.4 Nobody migrates. sFLR is integrated in every venue on Flare, has a liquid market, and can offer above-router yield by self-validating. eFLR launches with a lower yield and no integrations. Capital does not move for safety it has not been made to feel. Under Scope B alone this is likely; the Cosmos LSM's 25% cap was never binding because private ATOM liquid staking never reached it. Under Scope C the question is moot for concentration (the pool routes through the protocol either way) and open for custody. Test: eighteen months after launch, eFLR below 10% of liquid-staked FLR under Scope B is a failed launch; under Scope C the metric is share of pooled stake routed, and below 80% means the threshold in 10.4 was set too high.

11.5 The settlement gap is not closed. If neither FCC-managed keys nor a P-chain primitive ships, Scope B is a private LST run by the Foundation, with the same four keys under a different name. In that case ship Scope A only, and say so.

11.6 Liquidity crisis. A run on eFLR during a market drawdown empties the reserve, the exit queue extends to the longest outstanding lock (up to a year), and eFLR trades at a discount on secondary markets. This is not a failure of the design; it is what a liquid token over locked stake does under stress, and every LST has done it. The failure is if the discount is used as an argument for administrator intervention in the reserve. The invariant in 5.3 exists to make that intervention impossible, and the document should expect the demand for it.

11.7 The 20% floor collapses to the floor. Section 7.4: the router makes 20% the ceiling. If the network's intention in FIP.16 was to allow entities to price above the floor, the router contradicts it. Governance should decide this on purpose.

11.8 Ethereum was right. The strongest external argument is that enshrinement was rejected on the network with the most research behind it, and that the rejection was on principle: a protocol that selects validators has taken a political role. Flare's answer is that it took that role in FIP.16 already and that its validator set is small enough to make an algorithmic router auditable. If the Foundation does not accept the first half of that answer, this RFP has no ground to stand on, and the right document is a FIP.02 amendment, not this one.

11.9 The router's inputs are wrong. The router allocates on FSP reward data and uptime. If the FTSO layer is compromised, the router delegates toward whoever is gaming it, at scale, every epoch. Two limits on that. A compromised FTSO is a compromised network; the router is not the largest thing that breaks, and a document that solved it here would be overclaiming. And the router can be made to fail closed: if any entity's inputs move more than a bound in one epoch, or the network median moves more than a bound, the router issues no new instructions that epoch and existing delegations mature on schedule. A router that pauses is a delegator that stops, which is what Sceptre's four keys could not be made to do.

Cross-checking against a second data source is desirable and, for host attribution, not yet oracle-grade (Open Question 2); the fail-closed rule does not depend on it.

11.10 Regulators read "protocol-issued yield-bearing token" and see a fund. eFLR is a claim on a pool managed by a published algorithm with a fee. Under MiCA and under most securities analyses that description is close enough to a collective investment scheme to require an answer, and the answer "there is no issuer" is weaker for a Foundation-deployed contract than for a private one. Scope A carries none of this; it issues nothing. The RFP's position is that a written legal analysis for the EU, UK and US precedes any Scope B deployment on Flare, and that its absence is a reason to ship A alone, not a reason to ship B and hope.

11.11 The enclave is broken. Under §5.2 (i) the keys live in an FCC image. If the image is compromised and the keys extracted, the P-chain delegations can be redirected as they mature and the reserve can be drained. There is no fund that covers this (§7.2), the loss falls on eFLR holders, and recovery is a hard fork or nothing. This is the argument for the primitive over the enclave, stated as a failure mode so that it is not lost: a route whose failure is total is a bridge, not a destination. Under (ii) the equivalent failure is a consensus bug in the primitive, which is why B waits for two Songbird cycles.

12. What Ships First

A percentage cap per entity in FIP.02/FIP.05, which needs no part of this document. Then Scope A on Songbird: the router, mandatory above a per-pool threshold, without protocol custody. Then, on the Stage 3 consensus change, Scope B with custody under the envelope in Section 8, or an explicit decision not to. The order matters because A bounds the pool that exists today and B does not exist until the settlement gap is closed.

The one thing this RFP asks the network not to do is to treat the Sceptre case as closed when the second identity is deregistered on 16 September. The identity was never the problem. The 2.4 billion is still there, still uncapped, and the next pool to reach the ceiling will do the arithmetic Sceptre did.

Appendix A — Minimum Specification for the Settlement Bridge

Written from outside the codebase, as the questions a core reviewer would have to answer for each route. Where a mechanism is named, it is the Avalanche-derived mechanism Flare's go-flare inherits; where it is not, the RFP does not know.

A.1 P-chain primitive (§5.2 ii). A new P-chain transaction type, or an extension of AddPermissionlessDelegatorTx, whose reward owner and change owner are a C-chain

contract address rather than a P-chain key, and which the P-chain accepts only when accompanied by a proof that the named contract emitted a matching intent (entity, nodes, amount, duration) in a finalised C-chain block. The proof can follow the pattern `PChainStakeMirror` already uses in the other direction. Intents are emitted only at reward-epoch boundaries, batched per epoch by the router, so the transaction count is bounded by the entity count, not by depositor activity; a contract cannot flood the P-chain because the contract only speaks once per epoch. C-chain reorgs are not a concern past finality, and the P-chain acts only on finalised intents. At maturity the P-chain returns principal and rewards to the contract's atomic-memory address, and the router imports it in the next epoch. What the core team would need to state: whether the atomic export/import path can be driven without a P-chain signer at all, whether the mirror's verifier set is the right trust base for the reverse proof, and what the Stage 3 consensus changes do to any of this.

A.2 FCC enclave (§5.2 i). A TEE image that holds one P-chain key pair, generated inside the enclave, with remote attestation of the image hash published on-chain and checked by the eFLR contract before any instruction is honoured. The image executes one policy: read the router's target allocation from a finalised C-chain block, construct the export, delegate and import transactions that move toward it, and sign nothing else. The RFP would need the Foundation to state which TEE (the FCC roadmap will decide this, and the answer determines the side-channel history the design inherits), who can publish a new image hash and under what delay, and what happens to in-flight delegations when an image is retired: the honest answer is that they mature under the old key and the new image must be able to import them, which means the key has to be transferable between images or the retirement has to wait a full lock ladder. That constraint alone is an argument for A.1.

A.3 Either route. The router's target allocation is the only input the bridge accepts, the bridge's only outputs are delegations and imports, and both are visible on-chain every epoch. Anyone can recompute the allocation from public inputs and compare. A bridge whose behaviour cannot be recomputed by an outsider is a key, whatever it is called.

13. Sources

- FIP.05 — Update Services, Limits, and Rewards Required for Staking (accepted 8 Sep 2023): node/entity limits, delegation factor, uptime, "not enforced by the P-chain staking mechanism but by the mirroring service ... until phase 3". proposals.flare.network/FIP/FIP_5.html
- FIP.16 — Restructure FLR Tokenomics for Long-Term Network Sustainability (accepted 24 Apr 2026): inflation 5%→3%, FIRE, single-builder MEV roadmap (Stages 1–3), 5× P-chain signing weight, node cap 200M→300M, 20% minimum entity fee, FIRE negative governance. proposals.flare.network/FIP/FIP_16.html
- FIP.02 — FTSO Management Group (chill / second-strike ban). proposals.flare.network/FIP/FIP_2.html

- Flare, Staking Phase 2 (Nov 2023); Flare Developer Hub, FLR staking documentation (5% vote-power reward cap per validator)
- Kiln, Flare (FLR) validator documentation: no slashing on P-chain; delegation lock 60–365 days; C-chain vs P-chain mechanics
- Forum proposal, Jon-Sn0w, forum.flare.network, 31 Aug 2026 (Rotko/Sceptre multiple identities; role grants 15 Jul 2026; withdrawal→self-bond sequences)
- Sceptre, "Sceptre response and remediation status", 4 Sep 2026; Sceptre, Monthly recap May 2026 (FIP.16 impact, own-validator plans)
- Flare Metrics, 5 Sep 2026: 21.51bn active stake, 179 validators, 8.6bn free delegation space, largest entity Bifrost 1.14bn
- Janus Dispatch, "The Ceiling on Owning" (6 Sep 2026): pool 2.384bn / 11.08%, fee-avoidance estimate ~30M FLR/yr, four-keys critical path, proposed contract bounds
- Janus Dispatch, "From Fee-Burn to Sovereign Capital" (19 Jul 2026): FIRE mandate ordering, autonomous-contract-in-FCC model, the four buckles
- Cosmos Hub, Liquid Staking Module (governance-approved 2023): 25% global liquid staking cap, validator bond factor 250
- Polkadot, Nomination Pools (Nov 2022): protocol-native pools, no liquid token
- Ethereum: V. Buterin, "Should Ethereum be okay with enshrinement?" (Sep 2023); "Possible futures of the Ethereum protocol, part 3: The Scourge" (Oct 2024); B. Monnot, "Unbundling staking: towards rainbow staking" (ethresear.ch, Feb 2024)
- Avalanche validator documentation: 2% minimum delegation fee, 5× delegation factor (for comparison with Flare's 20% / 15×)

Figures marked as author's estimates are derived in "The Ceiling on Owning" and reproduced without recomputation.

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